# ENGINEERING MATHEMATICS-II

Department of Science and Humanities

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# UNIT 1: INTEGRAL CALCULUS

## Class 7:applications of triple integral and problems on finding VOLUME USING TRIPLE INTEGRAL

## APPLICATIONS OF TRIPLE INTEGRAL

- The triple integral

$$V(S) = \iiint_S 1\, dV$$

represents the *volume* of the solid $S$.

- The *average value* of the function $f = f(x, y, x)$ over a solid domain $S$ is given by

$$f_{\mathrm{AVG}(S)} = \left(\frac{1}{V(S)}\right) \iiint_S f(x, y, z)\, dV,$$

where $V(S)$ is the volume of the solid $S$.

# APPLICATIONS OF TRIPLE INTEGRAL

- The *center of mass* of the solid $S$ with density $\delta = \delta(x, y, z)$ is $(\bar{x}, \bar{y}, \bar{z})$, where

$$\bar{x} = \frac{\iiint_S x\,\delta(x, y, z)\,dV}{M},$$

$$\bar{y} = \frac{\iiint_S y\,\delta(x, y, z)\,dV}{M},$$

$$\bar{z} = \frac{\iiint_S z\,\delta(x, y, z)\,dV}{M},$$

and $M = \iiint_S \delta(x, y, z)\,dV$ is the mass of the solid $S$.

# APPLICATIONS OF TRIPLE INTEGRAL

The **second moments** (or **moments of inertia**) about the and *x*-, *y*-, and *z*-axes are as follows.

$$I_x = \iiint_Q (y^2 + z^2)\rho(x, y, z)\, dV$$ Moment of inertia about $x$-axis

$$I_y = \iiint_Q (x^2 + z^2)\rho(x, y, z)\, dV$$ Moment of inertia about $y$-axis

$$I_z = \iiint_Q (x^2 + y^2)\rho(x, y, z)\, dV$$ Moment of inertia about $z$-axis

## VOLUME USING TRIPLE INTEGRAL: DEFINITION

The simplest application of triple integrals is to compute volumes in an alternate way.

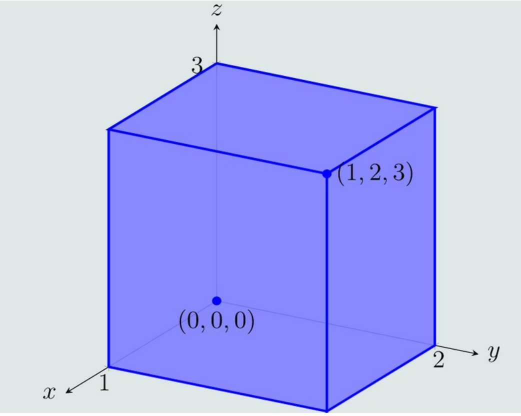
If the three-variable function $f$ is the constant 1, then the triple integral $\iiint_S dV$ evaluates to the volume of the closed bounded region $S$.

If the three-variable function $f$ is the constant 1 and $S$ is bounded by constants, then we are simply computing the volume of a rectangular box.

# VOLUME USING TRIPLE INTEGRAL: EXAMPLE:1

Compute the volume of the box with opposite corners at $(0,0,0)$ and $(1,2,3)$.

$$0 \le x \le 1,\ 0 \le y \le 2,\ 0 \le z \le 3.$$



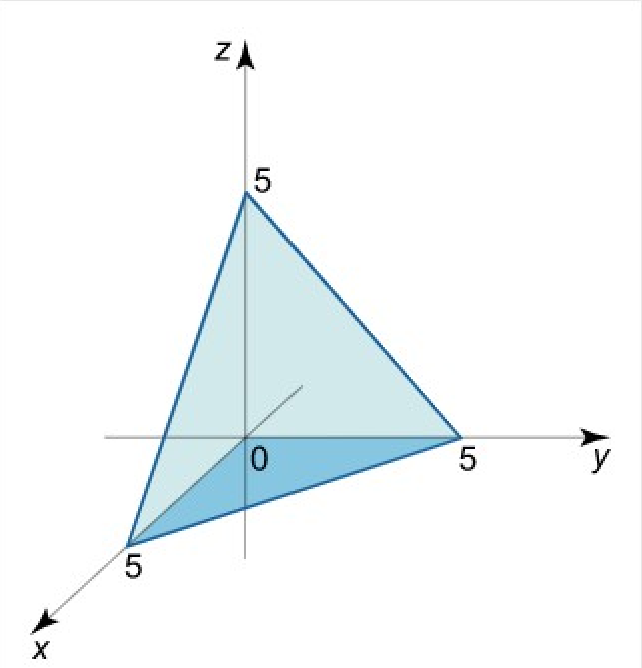

$$\begin{aligned}\int_0^1 \int_0^2 \int_0^3 dz\,dy\,dx &= \int_0^1 \int_0^2 z\Big|_0^3\,dy\,dx \\ &= \int_0^1 \int_0^2 3\,dy\,dx \\ &= \int_0^1 3y\Big|_0^2\,dx \\ &= \int_0^1 6\,dx = 6.\end{aligned}$$

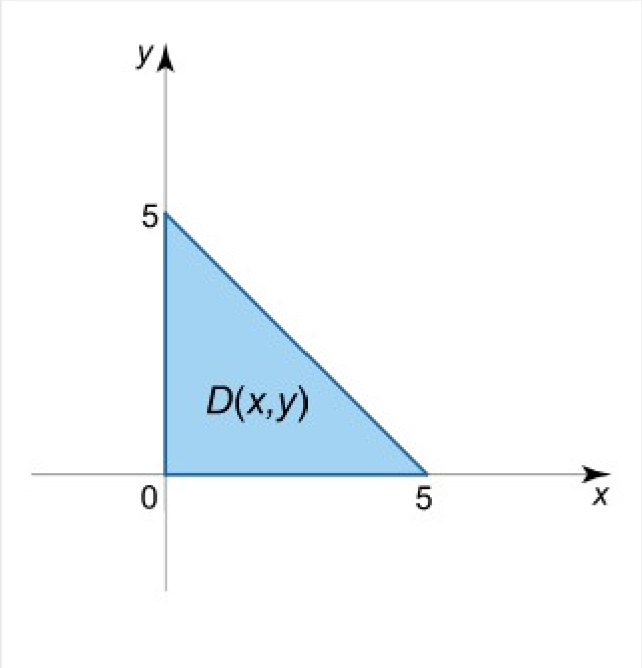
$$\int_0^3\int_0^2\int_0^1 dx\,dy\,dz,\quad \int_0^3\int_0^2\int_0^1 dx\,dy\,dz,\quad \int_0^3\int_0^1\int_0^2 dy\,dx\,dz,$$

$$\int_0^1\int_0^3\int_0^2 dy\,dz\,dx,\quad \int_0^2\int_0^1\int_0^3 dz\,dx\,dy, \text{ or } \int_0^1\int_0^2\int_0^3 dz\,dy\,dx.$$

## VOLUME USING TRIPLE INTEGRAL: EXAMPLE:2

Find the volume of the tetrahedron bounded by the planes $x + y + z = 5$, $x = 0$, $y = 0$,



# VOLUME USING TRIPLE INTEGRAL: EXAMPLE:2 contd…..

Representing the triple integral as an iterated integral, we can find the volume of the tetrahedron:

$$V = \iiint\limits_U dx\,dy\,dz = \int\limits_0^5 dx \int\limits_0^{5-x} dy \int\limits_0^{5-x-y} dz = \int\limits_0^5 dx \int\limits_0^{5-x} dy \cdot \left[ z|_0^{5-x-y} \right]$$

$$= \int\limits_0^5 dx \int\limits_0^{5-x} (5-x-y)\,dy = \int\limits_0^5 dx \left[ \left( 5y - xy - \frac{y^2}{2} \right) \Big|_{y=0}^{y=5-x} \right]$$

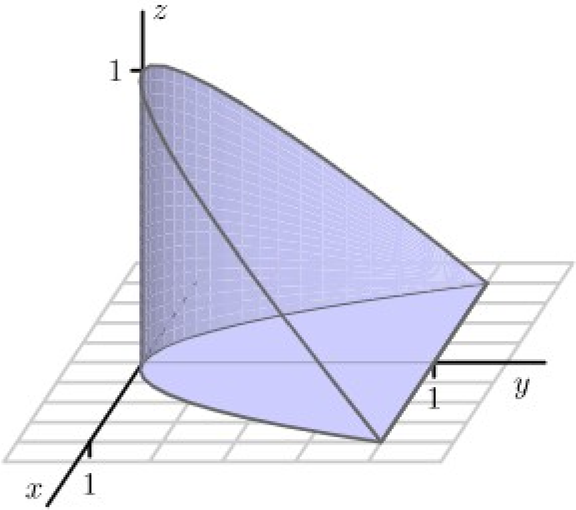
$$= \int\limits_0^5 \left[ 5(5-x) - x(5-x) - \frac{(5-x)^2}{2} \right] dx$$

$$= \int\limits_0^5 \left( 25 - 5x - 5x + x^2 - \frac{25 - 10x + x^2}{2} \right) dx = \frac{1}{2} \int\limits_0^5 \left( 25 - 10x + x^2 \right) dx$$

$$= \frac{1}{2} \left[ \left( 25x - \frac{10x^2}{2} + \frac{x^3}{3} \right) \Big|_0^5 \right] = \frac{1}{2} \left( 125 - 5 \cdot 25 + \frac{125}{3} \right) = \frac{125}{6}.$$

# VOLUME USING TRIPLE INTEGRAL: EXAMPLE:3

Consider the solid $S$ that is bounded by the parabolic cylinder $y = x^2$ and the planes $z = 0$ and $z = 1 - y$




$$z : 0 \rightarrow 1 - y$$

$$y : 0 \rightarrow x^2$$

$$x : -1 \rightarrow 1$$

# THANK YOU